

SIMATS ENGINEERING ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 10%

QUESTIONS: 2

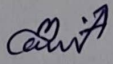
Duration : 45 Minutes
Total Marks:20

Annexure A

Descriptive Written Test

Code of Conduct:

I Priyanka G / 192524124 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: 

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Annexure A

Descriptive Written Test

1. A smart home automation system is being designed to manage lighting, temperature, and security based on user behavior and environmental conditions. The system must respond to real-time inputs such as motion detection, temperature changes, and time of day while also learning user preferences over time. In this context, explain the different types of intelligent agents (simple reflex, model-based, goal-based, and utility-based agents) and analyze how each type would function within this system. Justify which type of agent or hybrid approach would be most effective for ensuring comfort, energy efficiency, and security.
2. A robot is placed in an unknown maze and must find a path to the exit without prior knowledge of the layout. Explain how uninformed search strategies like Uniform Cost Search (UCS) and Iterative Deepening Search (IDS) can help solve this problem. Discuss the advantages and disadvantages of these algorithms in terms of time and space complexity. Relate the problem to different types of intelligent agents and explain how agent design affects decision-making. Suggest which combination of agent type and search strategy would be most effective and justify your choice.
3. A smart traffic management system is being designed to control traffic signals across a city based on vehicle density, pedestrian movement, and weather conditions. The system must respond to real-time sensor inputs while also learning traffic patterns over time to reduce congestion. In this context, explain the different types of intelligent agents (simple reflex, model-based, goal-based, and utility-based agents) and analyze how each type would function within this system. Justify which type of agent or hybrid approach would be most effective for minimizing congestion and travel time.

DATE: 25/07/26

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CSA1714 ARTIFICIAL INTELLIGENCE

CO 1 AT 3 DESCRIPTIVE TEST.

[UNIT 1: Introduction to AI]

1) SMART HOME APPLICATION SYSTEM USING INTELLIGENT AGENTS

Introduction:

A smart home automation system controls lighting, temperature and security by using sensors such as motion detector, temperature sensors, cameras, and time-based inputs.

Intelligent agents enable the system to make decision automatically and improve user comfort, energy efficiency and security.

(i) Simple Reflex Agent:

A simple reflex agent makes decisions only based on the current percept without considering previous information.

Working:

- Turn ON lights when motion detected.
- Turn OFF lights when no motion detected.
- Activate alarm when an unauthorized person is detected.

Advantages:

Fast response and simple implementation

Disadvantage:

Can't learn user habits and can't handle incomplete information.

(ii) Model-Based Reflex Agent

A model based reflex agent maintains an internal model of the environment using previous observations.

Working:

- Remembers whether rooms are occupied
- Adjusts AC according to previous temperature readings.
- Detects abnormal situations using stored environment info.

Advantages:

- Works in partially observed environments.
- Better decision-making than simple reflex agents.

Disadvantages:

- Requires memory.
- Slightly higher computational cost.

(ii) Goal-Based Agent

A Goal based agent selects actions that helps achieving predefined goals.

Goals in smart Home:

- maintain room temperature at 24°C .
- Ensure home security & Reduce electricity usage.

Working:

- chooses whether to turn ON fan or AC.
- Plans security actions when suspicious activity occurs.
- Decides the best sequence of actions to reach the desired goal.

Advantage:

Flexible and can evaluate multiple possible actions.

(iv) Utility-Based Agent

A utility-based agent chooses actions that maximize overall satisfaction (utility).

Working: It balances multiple objectives such as:

- Comfort
- Energy Efficiency
- Security
- Electricity Cost

Example: Instead of cooling the room to 18°C , it selects 24°C because it provides comfort while reducing power consumption.

Advantages:

makes optimal decisions & handles conflicting objectives efficiently.

Disadvantage:

Designing the utilities function is complex.

→ Hybrid Approach

The best solution is a **Hybrid Agent** combining:

Simple Reflex Agent for immediate responses.

Model-Based Reflex Agent for remembering environmental conditions.

Goal-Based Agent for planning.

Utility-Based Agent for optimizing comfort, security & energy usage.

→ Justification:

The hybrid system responds quickly to emergencies, let the learning of user preferences where minimizes electricity consumption and maximizes comfort, security which also adapts to changing environment.

Conclusion: A Hybrid - Utility based Intelligent Agent is most suitable.

(ii) Iterative Deepening Search (IDS)

Working: IDS repeatedly performs Depth-Limited Search with increasing depth limits.

Depth Limits: Level 0 $\rightarrow$ Level 1 $\rightarrow$ Level 2 $\rightarrow$ Continue until exit found.

Advantages:

- Complete and Uses very little memory
- Finds the shallowest solution.

Disadvantages:

- Repeats exploration of upper levels.
- Inefficient when path costs differ.

Complexity:

- Time: $O(b^d)$
- Space: $O(bd)$

where: b = branching factor, d = depth of solution

$\Rightarrow$ Comparison:

Feature	UCS	IDS
Complete	Yes	Yes
Optimal	Yes	only if equal step costs
Memory	High	Low
Time	High	Moderate
Suitable for path (weight)	Yes	No

Intelligent Agent Relation.

» Simple Reflex Agent:

Makes immediate movement decision but cannot remember visited paths.

» Model-Based Agent

Stores explored locations, avoids revisited dead ends and builds an internal map.

» Goal-Based Agent

Plans actions to reach exit, uses search algorithm effectively.

» Utility-Based Agent

Chooses paths with minimum energy, distance and time.

→ Recommended Combination - Hybrid System:

Model Based Agent and Uniform Cost Search and Goal Based Agent.

→ Justification:

- The model based agent remembers explored paths.
- The goal based agent plans toward the exit,
- UCS guarantees the least cost-path.
- Produces efficient navigation even in unknown environments.

Conclusion:

A Model-Based Goal-Based Agent with Uniform cost search provides the best solution because it remembers the environment, plans intelligently and guarantees the optimal path while adapting to an unknown maze.